

Sales Analytics Report

FutureInterns Task 1 - Superstore Sales Analysis

Python • Pandas • Matplotlib

Dataset	Superstore Sales Dataset
Records	9,994
Columns	22
Analysis Period	2014-01-03 to 2017-12-30
Total Sales	\$2,297,200.86
Total Profit	\$286,397.02
Profit Margin	12.47%

Prepared as part of FutureInterns Task 1

1. Executive Summary

This report analyzes the Superstore sales dataset to understand overall sales performance, profitability, regional performance, product performance, monthly sales patterns and the relationship between discounts and profit.

The dataset contains 9,994 sales records across 21 columns. The analysis was performed using Python and Pandas, while Matplotlib was used to create the visualizations.

KPI	Value
Total Sales	\$2,297,200.86
Total Profit	\$286,397.02
Total Quantity Sold	37,873
Average Sales per Record	\$229.86
Average Profit per Record	\$28.66
Profit Margin	12.47%

Overall Performance

- The dataset generated total sales of **\$2,297,200.86** and total profit of **\$286,397.02**.
- The overall profit margin is **12.47%**.
- A total of **37,873** units were sold across 9,994 transaction records.
- The average sales value per record is **\$229.86**.
- The highest monthly sales occurred in **2017-11**, with sales of **\$118,447.82**.
- The lowest monthly sales occurred in **2014-02**, with sales of **\$4,519.89**.

2. Data Preparation and Quality

Before performing the analysis, the dataset was checked for missing values, duplicate records and correct data types.

Check	Result
Rows before cleaning	9,994
Rows after cleaning	9,994
Columns	22
Duplicate rows	0
Missing values	0
Order Date	Converted to datetime
Ship Date	Converted to datetime

Data Quality Findings

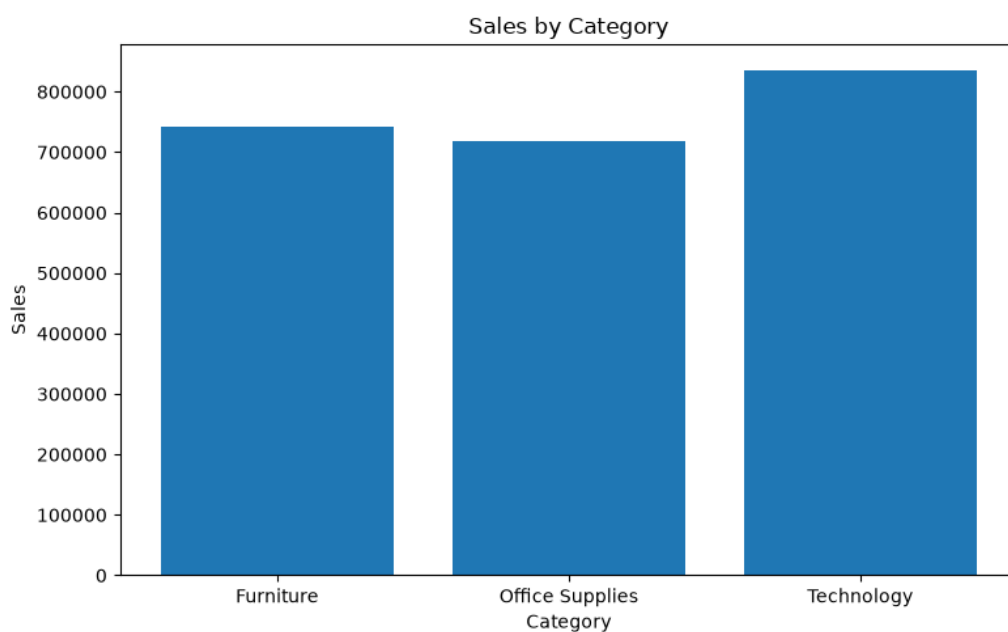
- All 9,994 records contain complete values in the provided dataset.
- No duplicate rows were found.
- Order Date and Ship Date were converted from text to datetime format.
- Sales, Quantity, Discount and Profit were retained as numerical fields.
- The dataset is suitable for sales, profit, product and regional analysis.

3. Category Performance

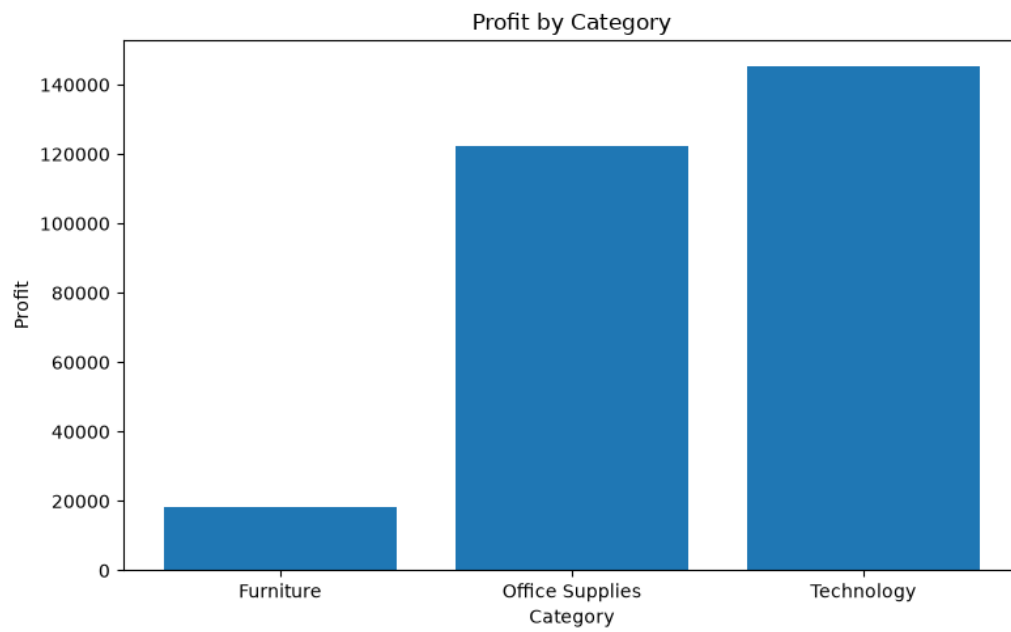
Sales and profit were aggregated by product category to compare the contribution of Furniture, Office Supplies and Technology.

Category	Sales	Profit
Technology	\$836,154.03	\$145,454.95
Furniture	\$741,999.80	\$18,451.27
Office Supplies	\$719,047.03	\$122,490.80

Sales by Category



Profit by Category

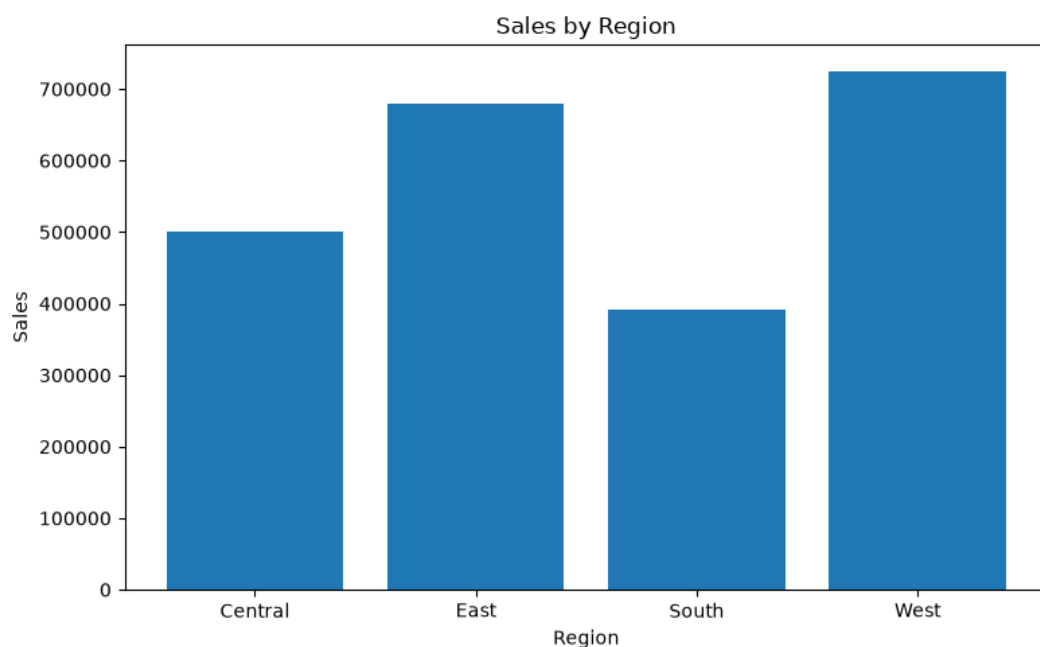


4. Regional Performance

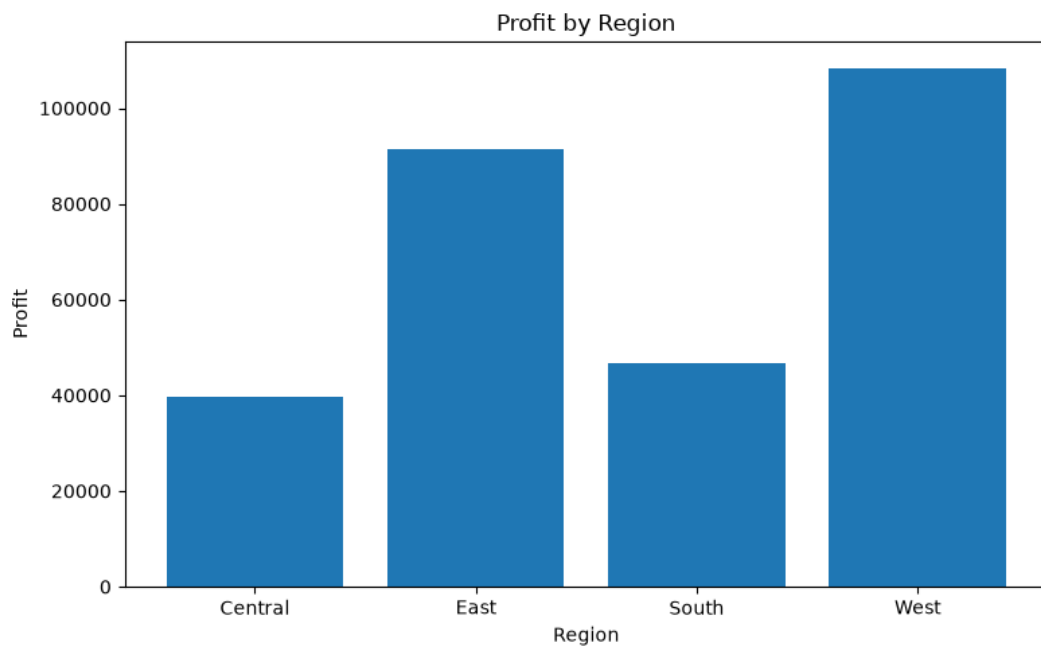
Regional analysis helps identify differences in sales and profit across the four geographic regions in the dataset.

Region	Sales	Profit
West	\$725,457.82	\$108,418.45
East	\$678,781.24	\$91,522.78
Central	\$501,239.89	\$39,706.36
South	\$391,721.91	\$46,749.43

Sales by Region

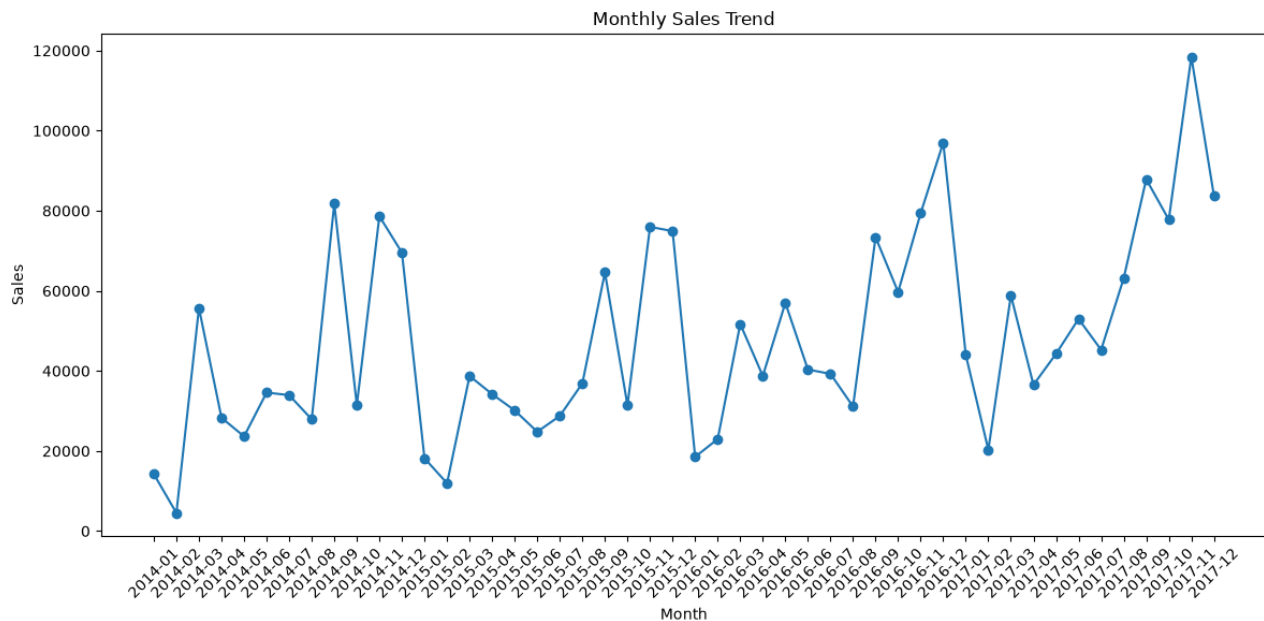


Profit by Region



5. Monthly Sales Trend

Monthly sales were calculated using Order Date. The trend helps identify periods of relatively high and low sales activity.



Metric	Month	Sales
Highest Monthly Sales	2017-11	\$118,447.82
Lowest Monthly Sales	2014-02	\$4,519.89

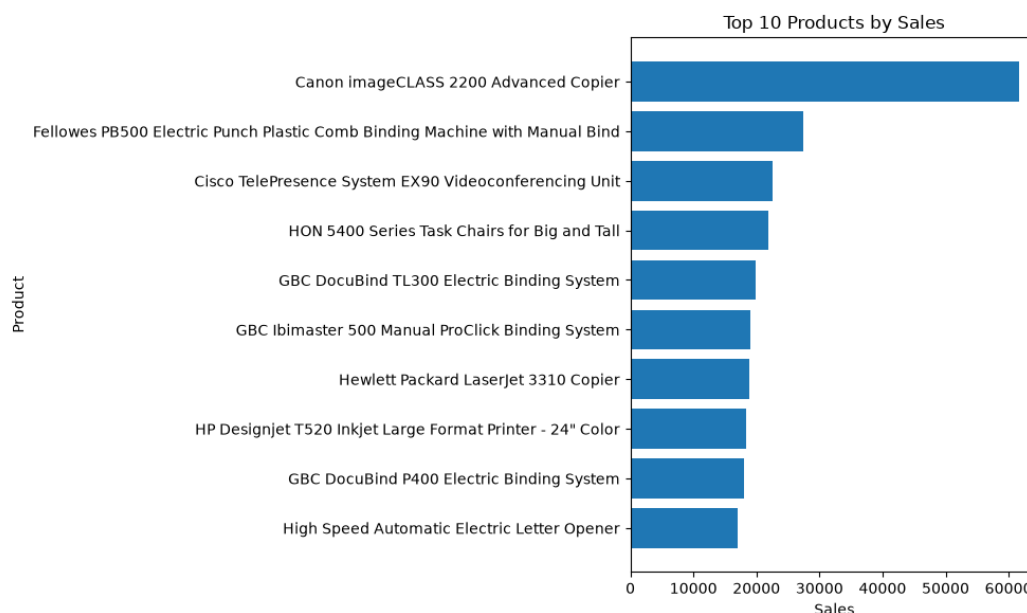
The highest monthly sales were recorded in 2017-11, while 2014-02 recorded the lowest monthly sales within the dataset.

6. Top Performing Products

Product-level analysis was performed using Sales, Profit and Quantity to identify products with strong contribution across different performance measures.

Top 10 Products by Sales

Rank	Product	Sales
1	Canon imageCLASS 2200 Advanced Copier	\$61,599.82
2	Fellowes PB500 Electric Punch Plastic Comb Binding Machine with Manual Bind	\$27,453.38
3	Cisco TelePresence System EX90 Videoconferencing Unit	\$22,638.48
4	HON 5400 Series Task Chairs for Big and Tall	\$21,870.58
5	GBC DocuBind TL300 Electric Binding System	\$19,823.48
6	GBC Ibimaster 500 Manual ProClick Binding System	\$19,024.50
7	Hewlett Packard LaserJet 3310 Copier	\$18,839.69
8	HP Designjet T520 Inkjet Large Format Printer - 24" Color	\$18,374.90
9	GBC DocuBind P400 Electric Binding System	\$17,965.07
10	High Speed Automatic Electric Letter Opener	\$17,030.31



7. Product Profitability and Quantity

Sales volume does not always represent the same performance as profitability. The following tables show the leading products by profit and quantity sold.

Top 10 Products by Profit

Rank	Product	Profit
1	Canon imageCLASS 2200 Advanced Copier	\$25,199.93
2	Fellowes PB500 Electric Punch Plastic Comb Binding Machine with Manual Bind	\$7,753.04
3	Hewlett Packard LaserJet 3310 Copier	\$6,983.88
4	Canon PC1060 Personal Laser Copier	\$4,570.93
5	HP Designjet T520 Inkjet Large Format Printer - 24" Color	\$4,094.98
6	Ativa V4110MDD Micro-Cut Shredder	\$3,772.95
7	3D Systems Cube Printer, 2nd Generation, Magenta	\$3,717.97
8	Plantronics Savi W720 Multi-Device Wireless Headset System	\$3,696.28
9	Ibico EPK-21 Electric Binding System	\$3,345.28
10	Zebra ZM400 Thermal Label Printer	\$3,343.54

Top 10 Products by Quantity Sold

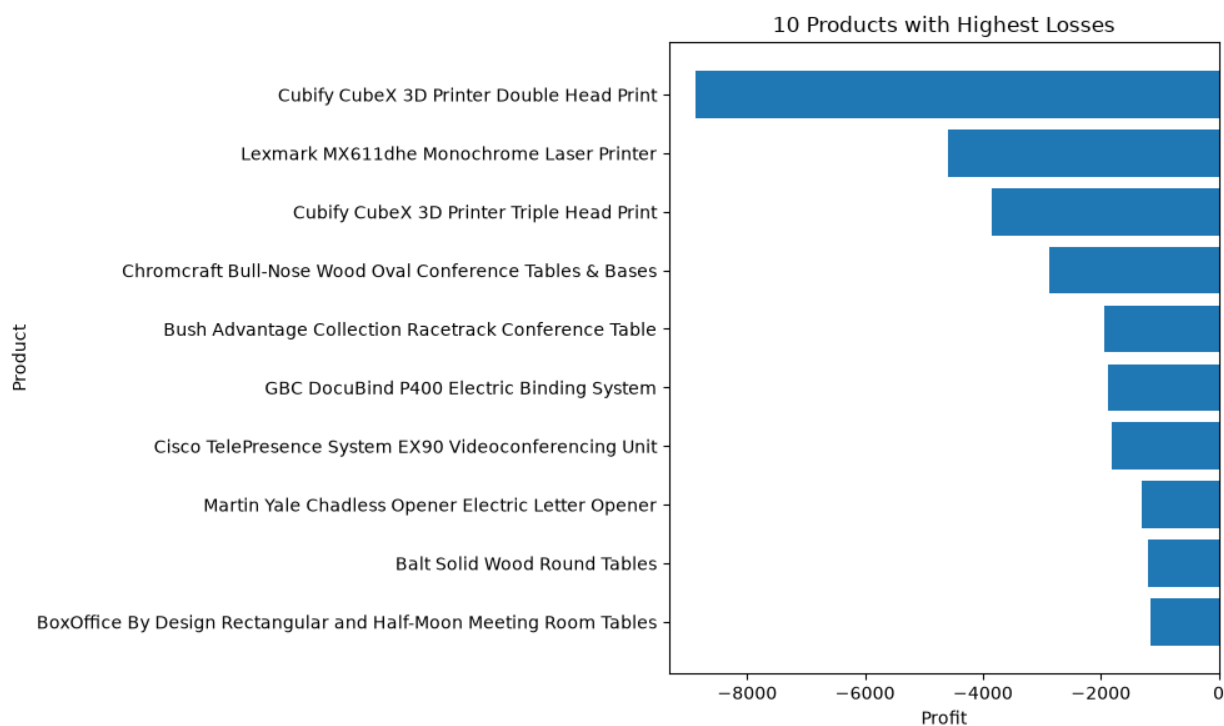
Rank	Product	Quantity
1	Staples	215
2	Staple envelope	170
3	Easy-staple paper	150
4	Staples in misc. colors	86
5	KI Adjustable-Height Table	74
6	Storex Dura Pro Binders	71
7	Avery Non-Stick Binders	71
8	GBC Premium Transparent Covers with Diagonal Lined Pattern	67
9	Situations Contoured Folding Chairs, 4/Set	64
10	Staple-based wall hangings	62

8. Loss-Making Products

The analysis identified **301** products with negative total profit. The following table presents the 10 products with the largest cumulative losses.

Rank	Product	Profit
1	Cubify CubeX 3D Printer Double Head Print	\$-8,879.97
2	Lexmark MX611dhe Monochrome Laser Printer	\$-4,589.97
3	Cubify CubeX 3D Printer Triple Head Print	\$-3,839.99
4	Chromcraft Bull-Nose Wood Oval Conference Tables & Bases	\$-2,876.12
5	Bush Advantage Collection Racetrack Conference Table	\$-1,934.40
6	GBC DocuBind P400 Electric Binding System	\$-1,878.17
7	Cisco TelePresence System EX90 Videoconferencing Unit	\$-1,811.08
8	Martin Yale Chadless Opener Electric Letter Opener	\$-1,299.18
9	Balt Solid Wood Round Tables	\$-1,201.06
10	BoxOffice By Design Rectangular and Half-Moon Meeting Room Tables	\$-1,148.44

Loss-Making Products Chart

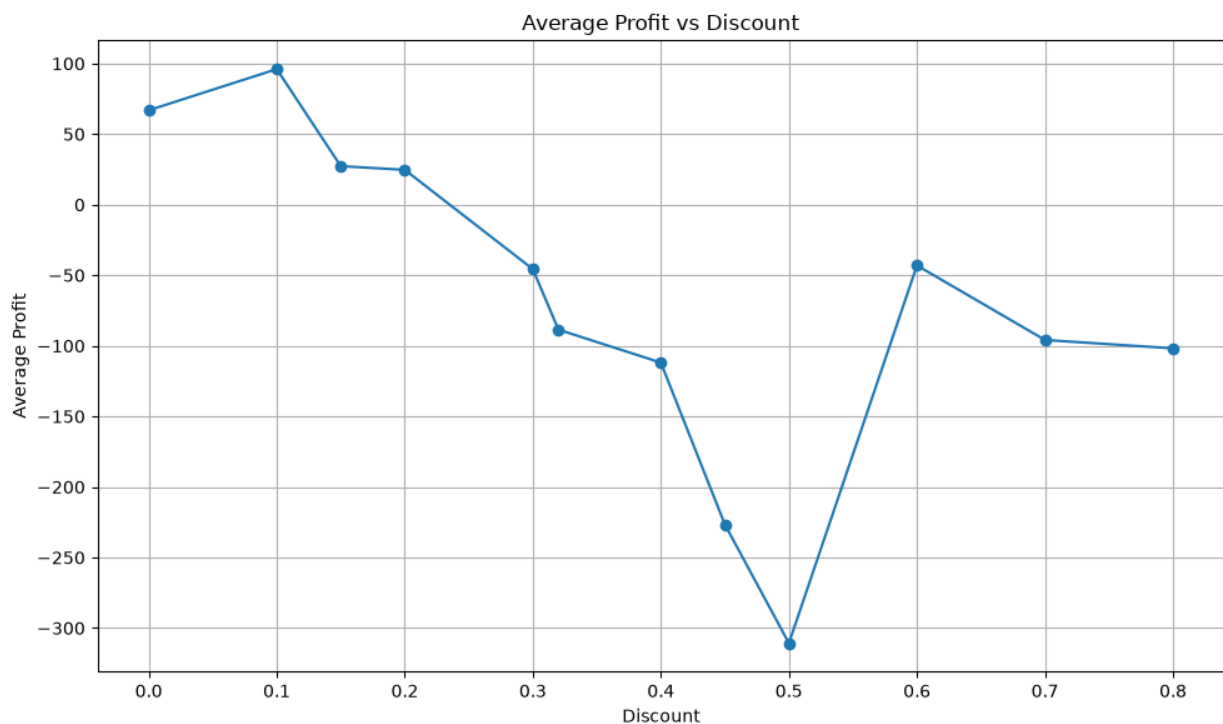


The largest product-level loss in the dataset was **\$-8,879.97** for **Cubify CubeX 3D Printer Double Head Print**.

9. Discount and Profit Analysis

Average profit was calculated for each discount level to understand how discount levels are associated with profitability.

Discount	Average Profit
0%	\$66.90
10%	\$96.06
15%	\$27.29
20%	\$24.70
30%	\$-45.68
32%	\$-88.56
40%	\$-111.93
45%	\$-226.65
50%	\$-310.70
60%	\$-43.08
70%	\$-95.87
80%	\$-101.80



The analysis shows that several higher discount levels are associated with negative average profit. This indicates that discount strategy should be monitored together with product margin and sales volume.

10. Key Insights

Overall Performance: Total sales were \$2,297,200.86 with total profit of \$286,397.02, resulting in an overall profit margin of 12.47%.

Category Performance: Technology generated the highest total sales at \$836,154.03.

Category Profitability: Technology generated the highest total profit at \$145,454.95.

Regional Sales: West recorded the highest total sales at \$725,457.82.

Regional Profit: West generated the highest total profit at \$108,418.45.

Monthly Trend: The highest monthly sales occurred in 2017-11, with \$118,447.82 in sales.

Product Performance: Canon imageCLASS 2200 Advanced Copier was the highest-selling product by sales value, generating \$61,599.82.

Product Profitability: Canon imageCLASS 2200 Advanced Copier generated the highest total product profit of \$25,199.93.

Loss-Making Products: 301 products recorded negative total profit. The largest loss was associated with Cubify CubeX 3D Printer Double Head Print.

Discount Effect: Higher discount levels in the dataset are frequently associated with lower or negative average profit.

11. Business Recommendations

Monitor High-Discount Transactions

Review products and transactions with high discounts because some discount levels are associated with negative average profit.

Review Loss-Making Products

Investigate pricing, discounts, shipping costs and demand for products with consistently negative profit.

Focus on Profitable Categories

Use category-level profit analysis when planning product promotion and inventory strategies.

Monitor Regional Performance

Compare sales and profit by region regularly to identify differences in regional performance.

Use Monthly Trends

Use historical monthly sales patterns to support inventory planning and promotional scheduling.

Separate Sales Volume from Profitability

A product with high quantity or high sales may not necessarily generate the highest profit, so both measures should be monitored.

12. Methodology

1. Data Loading

The Superstore CSV dataset was loaded using Pandas.

2. Data Cleaning

Missing values and duplicate rows were checked. Date columns were converted into datetime format.

3. KPI Calculation

Total sales, total profit, total quantity, average sales, average profit and profit margin were calculated.

4. Category Analysis

Sales and profit were aggregated by product category.

5. Regional Analysis

Sales and profit were aggregated by region.

6. Time Analysis

Monthly sales were calculated using Order Date.

7. Product Analysis

Products were ranked by sales, profit and quantity.

8. Loss Analysis

Products with negative total profit were identified.

9. Discount Analysis

Average profit was calculated for each discount level.

10. Visualization

Matplotlib charts were generated to communicate the analysis clearly.

Tools Used

Tool	Purpose
Python	Data analysis and automation
Pandas	Data cleaning and aggregation
Matplotlib	Data visualization
ReportLab	PDF report generation

End of Report